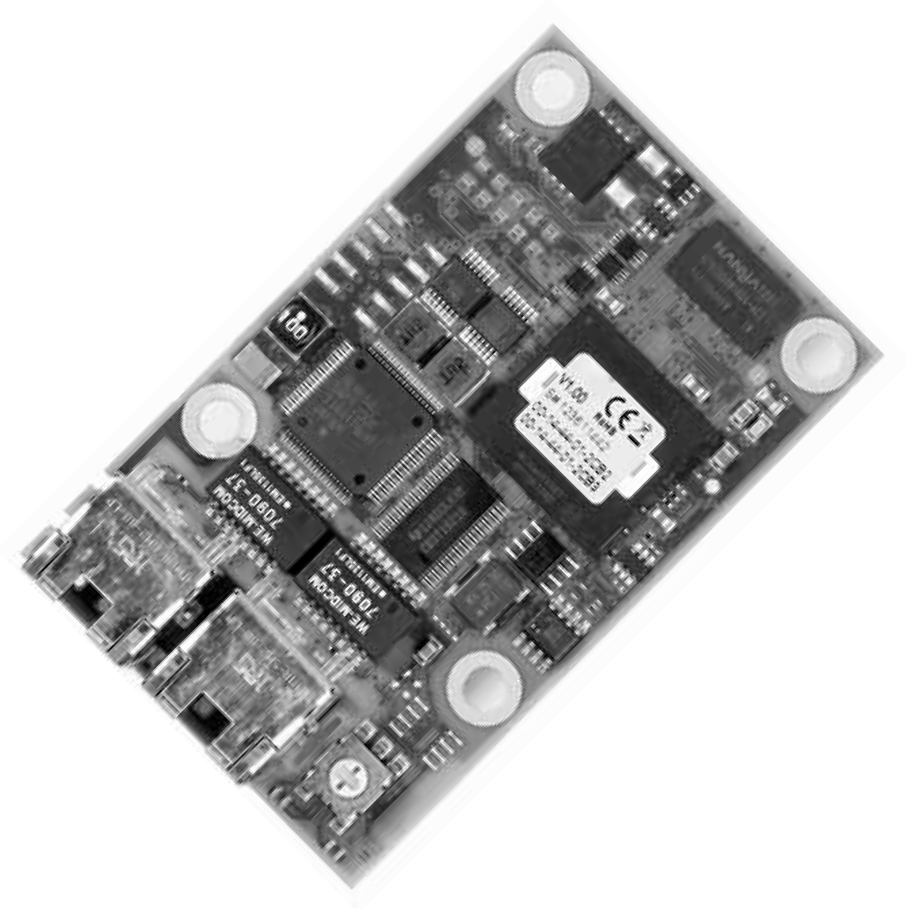
CIM 500 Ethernet module

Installation and operating instructions



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English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe the Grundfos CIM 500 Ethernet module.

Sections 1-3 provide the information necessary to be able to install and start up the product in a safe way.

Sections 4-8 provide important information about operating the product as well as information about fault finding, technical data and disposal of the product.

CONTENTS

Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information

1.1 Hazard statements

The symbols and hazard statements below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The hazard statements are structured in the following way:

SIGNAL WORD

Description of hazard

Consequence of ignoring the warning.

- Action to avoid the hazard.

1.2 Notes

The symbols and notes below may appear in Grundfos installation and operating instructions, safety instructions and service instructions.

Observe these instructions for explosion-proof products.

A blue or grey circle with a white graphical symbol indicates that an action must be taken.

A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.

If these instructions are not observed, it may result in malfunction or damage to the equipment.

Tips and advice that make the work easier.

2. Applications

The CIM 500 Ethernet module enables data transmission between an Industrial Ethernet network and a Grundfos product.

The module supports various Industrial Ethernet protocols. See section *3.3 Selection of Industrial Ethernet protocol*.

The module is fitted in the product to be communicated with, in a CIU 500 or in an E-box 500.

Retrofitting of the module is described in the installation and operating instructions of the Grundfos product.

Configuration is done via the built-in webserver, using a standard web browser on a PC.

You can download the specific functional profile for the product in question from Grundfos Product Center.

2.1 Abbreviations

APDU	Application Protocol Data Unit.
ARP	Address Resolution Protocol. Translates IP addresses to MAC addresses.
CAT5	Ethernet cable type with four twisted-pair cables.
CAT5e	Enhanced CAT5 cable with better performance.
CAT6	Ethernet cable compatible with CAT5 and CAT5e, and with very high performance.
CIM	Communication Interface Module.
CIU	Communication Interface Unit.
DHCP	Dynamic Host Configuration Protocol. Used to configure network devices so that they can communicate via an IP network.
DNS	Domain Name System. Used to resolve host names to IP addresses.
E-box	Extension Box. Used as a communication interface between a Grundfos product and a fieldbus.
GENIpro	Grundfos Electronics Network Intercommunication protocol Proprietary Grundfos fieldbus protocol.
GiC	Grundfos iSOLUTIONS Cloud
GND	Ground.
GRM	Grundfos Remote Management.
HTTP	Hyper Text Transfer Protocol. The protocol commonly used to navigate the world wide web.
IANA	Internet Assigned Numbers Authority.
IP	Internet Protocol.
LED	Light-emitting diode.
MAC	Media Access Control. Unique address for a piece of hardware.
MDI	Medium Dependent Interface.
Ping	Packet InterNet Groper. A software utility that tests connectivity between two TCP/IP hosts.
PLC	Programmable Logic Controller.
RJ45	Registered Jack #45, also called 8P8C modular connector type. Connects four twisted-pair cables. Most common Ethernet connector type.
SELV	Separated or Safety Extra-Low Voltage.
SELV-E	Separated or Safety Extra-Low Voltage with earth.
TCP	Transmission Control Protocol. Protocol for Internet communication and Industrial Ethernet communication.
UDP	User Datagram Protocol
URL	Uniform Resource Locator. The IP address used to connect to a server.
VPN	Virtual Private Networks.

2.2 CIM 500 Ethernet module

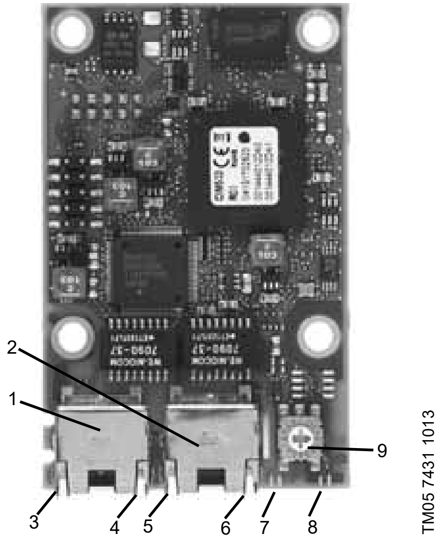

Fig. 1 CIM 500 Ethernet module

Pos.	Designation	Description
1	ETH1	Industrial Ethernet RJ45 connector 1
2	ETH2	Industrial Ethernet RJ45 connector 2
3	DATA1	Data activity LED for RJ45 connector 1
4	LINK1	Link LED for RJ45 connector 1
5	DATA2	Data activity LED for RJ45 connector 2
6	LINK2	Link LED for RJ45 connector 2
7	LED1	Red and green status LED for the selected Ethernet protocol
8	LED2	Red and green LED for internal communication between CIM 500 and the Grundfos product
9	SW1	Rotary switch for selection of Industrial Ethernet protocol

3. Installation

WARNING

Electric shock

Death or serious personal injury

- Only connect CIM 500 to SELV or SELV-E circuits.

Fig. 2 QR code for the CIU quick guide

3.1 Security

Grundfos connected products must be behind a firewall or connected to a private network.

If a firewall or private network is not in place, the Grundfos product may be subject to a cyber-security risk and becomes vulnerable to an attack or compromise.

Follow the highly recommended configuration requirements below. If in doubt, consult an IT Infrastructure Specialist.

3.1.1 CIM 500

CIM 500 is a traditional network-connected device and must be placed on a private network behind a firewall. It must not be connected directly to the Internet. Also, no TCP/IP ports must be forwarded to the product. If you need remote access to the device, you must use technologies such as Virtual Private Networks (VPNs) to ensure a secured connection. Consider contacting an IT Infrastructure Specialist to establish such a solution.

For CIM 500 in mode 4, GRM IP (Grundfos Remote Management) the firewall must accept connections initialised by the Grundfos product to the Internet only (outgoing connections).

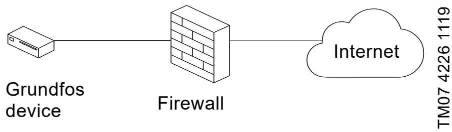

Fig. 3 Secure connectivity for CIM 500

3.2 Connecting the Ethernet cable

Use RJ45 plugs and an Ethernet cable. Connect the cable screen to protective earth at both ends.

It is important to connect the cable screen to protective earth by means of the earth clamp or in the connector.

For GRM IP communication, ensure that the network is protected by a firewall.

Maximum cable length

Speed [Mbit/s]	Cable type	Max. cable length [m/ft]
100/10	CAT5, CAT5e, CAT6	100/328

CIM 500 is designed for flexible network installation. The built-in two-port switch makes it possible to daisy chain from product to product without the need of additional Ethernet switches. The last product in the chain is only connected to one of the Ethernet ports. Each Ethernet port has its own MAC address and CIM 500 has a built-in switch which means that the cable can run another 100 metres whenever passing a CIM 500 module.

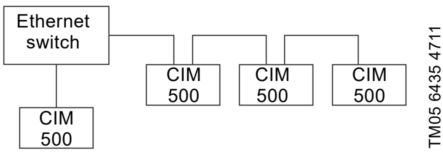

Fig. 4 Example of Industrial Ethernet network

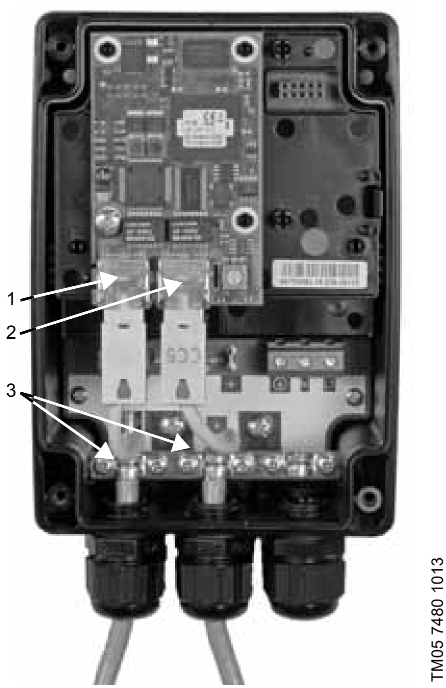

Fig. 5 Example of Ethernet connection

Pos.	Description
1	Industrial Ethernet RJ45 connector 1
2	Industrial Ethernet RJ45 connector 2
3	Earth clamp/GND

3.3 Selection of Industrial Ethernet protocol

The module has a rotary switch for selection of the Industrial Ethernet protocol. See fig. 6.

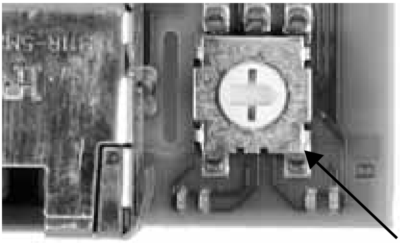

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Fig. 6 Selecting the Industrial Ethernet protocol

Pos.	Description
0	PROFINET IO, default
1	Modbus TCP
2	BACnet IP
3	EtherNet/IP
4	GRM IP for Grundfos Remote Management, requires a contract with Grundfos.
5	Grundfos iSOLUTIONS Cloud (GiC)
6...E	Reserved. LED1 is permanently red to indicate an invalid configuration.
F	Resetting to factory settings. 1. Set the rotary switch to this position 2. LED1 starts to flash red and green for 20 seconds to indicate that factory resetting is about to take place. 3. After 20 seconds, LED1 stops to flash and factory resetting is initiated. 4. When both LED1 and LED2 switch off, the resetting is completed. The rotary switch can be moved to another position.

If the rotary switch position is changed when the module is powered on, the module will restart and use the protocol associated with the new position.

3.4 Setting up the IP addresses

The module comes with a fixed webserver IP address. Via the webserver, this address can be changed to another fixed value, or a DHCP server can be selected.

Default IP settings used by webserver	IP address: 192.168.1.100 Subnet mask: 255.255.255.0 Gateway: 192.168.1.1
Device name and IP settings for PROFINET IO	The device name is configured from the webserver or from the PROFINET IO configuration tool. The IP address is automatically assigned by the PLC. This assigned PROFINET IP address must be different from the webserver's IP address.
IP settings for Modbus TCP	It can be given a fixed value via the webserver, or it can use a DHCP server. This assigned Modbus TCP address must be different from the webserver's IP address.
IP settings for BACnet IP	It can be given a fixed value via the webserver, or it can use a DHCP server. Note that BACnet IP and the webserver share the same IP address.
IP settings for EtherNet/IP	It can be given a fixed value via the webserver, or it can use a DHCP server. This assigned EtherNet/IP address must be different from the webserver's IP address.
IP settings for GRM IP	GRM IP uses DHCP, and the connected router automatically assigns an IP address.

3.5 Connection to the webserver

The module can be configured by means of the built-in webserver. To establish a connection from a PC to CIM 500, proceed as follows:

1. Connect the PC and the module using an Ethernet cable. See fig. 7.
2. Configure the Ethernet port of the PC to belong to the same subnetwork as CIM 500, for example 192.168.1.101, and the subnet mask to 255.255.255.0. See section *How to configure an IP address on your PC using Windows 7* or *How to configure an IP address on your PC using Windows 10*.
3. Open a standard Internet browser and type 192.168.1.100 in the URL field.
4. Log in to the webserver:

User name	Default: admin
Password	Default: Grundfos

The first time you log in, you must set a unique password. Only a factory resetting of CIM 500 can reset the password to Grundfos.

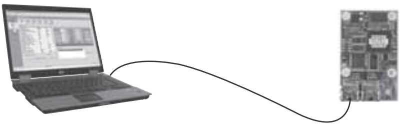

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Fig. 7 CIM 500 connected to PC via Ethernet cable

3.5.1 Connection to the webserver using GRM IP

To access the webserver via GRM IP, you need to connect your PC parallel to the router (RJ45 port 1) at CIM 500 RJ45 port 2. Look at the router webserver which address the router assigns to the CIM 500. Then use this IP address for the webserver access.

See also sections *3.1 Security* and *Webserver configuration*.

You can use both ETH1 and ETH2 to establish a connection to the webserver.

You can access the webserver while the selected Industrial Ethernet protocol is active.

When you select "GRM IP", the connected router automatically assigns an IP address to the module via DHCP.

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Fig. 8 CIM 500 connected to PC and cellular router

4. Status LEDs

The module has two status LEDs, see fig. 1:

- LED1: Red and green status LED for Ethernet communication, fieldbus LED.
- LED2: Red and green status LED for communication between the module and the Grundfos product, GENI LED.

LED1, PROFINET IO

Status	Description
Off	The module is switched off.
Flashing green	Wink function. LED flashes 10 times when activated from the PROFINET master.
Permanently green	The module is in cyclic data exchange mode.
Flashing red (3 Hz, duty cycle 50 %)	Wrong or missing PROFINET IO configuration. See section *6.1 PROFINET IO*.
Pulsing red (0.3 Hz, duty cycle 10 %)	The device name and network settings are configured, but connection to master lost. See section *6.1 PROFINET IO*.
Permanently red	Product is not supported. See section *6.1 PROFINET IO*.
Permanently red and green	Error in firmware download. See section *6.1 PROFINET IO*.
Flashing red and green	Resetting to factory settings. After 20 seconds CIM 500 restarts.

LED1, Modbus TCP and BACnet IP

Status	Description
Off	No Modbus or BACnet communication, or switched off.
Flashing green	Modbus or BACnet communication active.
Permanently red	Module configuration fault. See section *6.2 Modbus TCP or BACnet IP*.
Permanently red and green	Error in firmware download. See section *6.2 Modbus TCP or BACnet IP*.
Flashing red and green	Resetting to factory settings. After 20 seconds CIM 500 restarts.

LED1, EtherNet/IP

Status	Description
Off	Ethernet Link is not active.
Permanently green	Ethernet Link is active, connection is established.
Flashing green	Ethernet Link is active, no connection is established.
Permanently red	Ethernet Link is active, IP address conflict is detected.
Flashing red	Ethernet Link is active, any connection is timed out.

LED1, GRM IP

Status	Description
Off	No GRM IP communication or switched off.
Flashing green	Startup period (120 seconds)
Permanently green	Communication with the GRM server is established.
Flashing green (0.3 Hz)	Data is transmitted to the GRM server.
Flashing red	• GRM communication lost, • no communication to the GRM server, or • product driver is not loaded from GRM.
Flashing red and green	Resetting to factory settings. After 20 seconds CIM 500 restarts.

LED2, all fieldbuses

Status	Description
Off	The module is switched off.
Flashing red	No internal communication between CIM 500 and the Grundfos product.
Permanently red	The module does not support the Grundfos product connected.
Permanently green	Internal communication between the module and the Grundfos product is OK.
Permanently red and green	Memory fault.

During startup, there is a delay of up to five seconds before the status of LED1 and LED2 is updated.

5. Data activity and link LEDs

The module has two connectivity LEDs related to each RJ45 connector. See fig. 1.

DATA1 and DATA2

These yellow LEDs indicate data traffic for the RJ45 connector in question.

Status	Description
Off	No data communication on RJ45 connector
Flashing	Data communication on RJ45 connector
On	Heavy network traffic

LINK1 and LINK2

These green LEDs show whether the Ethernet cable is properly connected to the RJ45 connector in question.

Status	Description
Off	No Ethernet link on RJ45 connector
On	Ethernet link on RJ45 connector is OK

6. Fault finding

6.1 PROFINET IO

You can detect faults in a module by observing the status of the two status LEDs. See the tables below.

CIM 500 fitted in a Grundfos product or CIM 500 fitted in a CIU 500

Ensure that SW1 is in position "0" to select PROFINET.

Fault (LED status)	Possible cause	Remedy
1. Both LEDs remain off when the power supply is connected.	a) The module is fitted incorrectly in the Grundfos product.	Check that the module is fitted and connected correctly.
	b) The module is defective.	Replace the module.
	c) CIU 500 is defective.	Replace CIU 500.
2. LED1 remains off.	a) SW1 is not set correctly.	Set the switch to "0".
3. LED2 is flashing red.	a) No internal communication between the module and the Grundfos product.	Check that the module is fitted correctly in the Grundfos product.
	b) No internal communication between CIU 500 and the Grundfos product.	• Check the cable connection between the Grundfos product and CIU 500. • Check that the individual conductors have been connected correctly, for example not reversed. • Check the power supply to the Grundfos product.
4. LED2 is permanently red.	a) The module does not support the connected Grundfos product.	Contact the nearest Grundfos company.
5. LED1 is permanently red.	a) The module does not support the connected Grundfos product.	Contact the nearest Grundfos company.
	b) SW1 is in illegal position.	Set the switch to "0".
6. LED1 is flashing red, 3 Hz.	a) Fault in the PROFINET IO configuration of the module.	• Restart CIM 500. Use the RESTART button on the webserver, or power cycle the product. See section *PROFINET IO configuration*. • Check that the PROFINET IO IP address configuration is correct. Check the device name in CIM 500 and PROFINET IO master. • Check that the right GSDML file is used.
7. LED1 is pulsing red, 0.3 Hz.	a) The connection to the master is lost.	• Check the cables. • Check that the master is running.
8. LED1 is permanently red and green at the same time.	a) Error in firmware download.	Use the webserver to download the firmware again. See section *Update* in the appendix.
9. LED2 is permanently red and green at the same time.	a) Memory fault.	Replace the module.

6.2 Modbus TCP or BACnet IP

You can detect faults in a module by observing the status of the two status LEDs. See the tables below.

CIM 500 fitted in a Grundfos product or CIM 500 fitted in a CIU 500

Ensure that SW1 is in position "1" if Modbus should be selected or in position "2" if BACnet should be selected.

Fault (LED status)	Possible cause	Remedy
1. Both LEDs remain off when the power supply is connected.	a) The module is fitted incorrectly in the Grundfos product.	Check that the module is fitted and connected correctly.
	b) The module is defective.	Replace the module.
	c) CIU 500 is defective.	Replace CIU 500.
2. LED2 is flashing red.	a) No internal communication between the module and the Grundfos product.	Check that the module is fitted correctly.
	b) No internal communication between the CIU 500 and the Grundfos product.	• Check the cable connection between the Grundfos product and CIU 500. • Check that the individual conductors have been connected correctly, for example not reversed. • Check the power supply to the Grundfos product.
3. LED2 is permanently red.	a) The module does not support the connected Grundfos product.	Contact the nearest Grundfos company.
4. LED1 is permanently red.	a) Fault in the Modbus configuration of the module.	• Check that SW1 is set to "1". • Check that the Modbus IP address configuration is correct. See section *Modbus TCP configuration*.
	b) Fault in the BACnet configuration of the module.	• Check that SW1 is set to "2". • Check that the BACnet IP address and UDP port number configuration is correct. See section *BACnet IP configuration*.
5. LED1 is permanently red and green at the same time.	a) Error in firmware download.	Use the webserver to download the firmware again. See section *Update* in the appendix.
6. LED2 is permanently red and green at the same time.	a) Memory fault.	Replace the module.

6.3 EtherNet/IP

You can detect faults in a module by observing the status of the two status LEDs. See the tables below.

CIM 500 fitted in a Grundfos product or CIM 500 fitted in a CIU 500

Ensure that SW1 is in position "3".

Fault (LED status)	Possible cause	Remedy
1. Both LEDs remain off when the power supply is connected.	a) The module is fitted incorrectly in the Grundfos product.	Check that the module is fitted and connected correctly.
	b) The module is defective.	Replace the module.
	c) CIU 500 is defective.	Replace CIU 500.
2. LED1 remains off.	a) SW1 is not set correctly.	Set the switch to "3".
3. LED2 is flashing red.	a) No internal communication between the module and the Grundfos product.	Check that the module is fitted correctly.
	b) No internal communication between the CIU 500 and the Grundfos product.	• Check the cable connection between the Grundfos product and CIU 500. • Check that the individual conductors have been connected correctly, for example not reversed. • Check the power supply to the Grundfos product.
4. LED2 is permanently red.	a) The module does not support the connected Grundfos product.	Contact the nearest Grundfos company.
5. LED1 is permanently red.	a) IP address conflict.	Check the IP address configuration.
	b) SW1 is in illegal position	Check that SW1 is set to "3".
6. LED1 is flashing red.	a) Connection time-out.	Verify the connection and communication between PLC and CIM 500.
7. LED1 is permanently red and green at the same time.	a) Error in firmware download.	Use the webserver to download the firmware again. See section *Update* in the appendix.
8. LED2 is permanently red and green at the same time.	a) Memory fault.	Replace the module.

6.4 GRM IP

You can detect faults in a module by observing the status of the two status LEDs. See the tables below.

CIM 500 fitted in a Grundfos product or CIM 500 fitted in a CIU 500*

Ensure that SW1 is in position "4" to select GRM IP.

Fault (LED status)	Possible cause	Remedy
1. Both LEDs remain off when the power supply is connected.	a) The module is fitted incorrectly in the Grundfos product.	Check that the module is fitted and connected correctly.
	b) The module is defective.	Replace the module.
	c) CIU 500 is defective.	Replace CIU 500.
2. LED1 remains off.	a) SW1 is not set correctly.	Set the switch to "4".
3. LED2 is flashing red.	a) No internal communication between the module and the Grundfos product.	Check that the module is fitted correctly in the Grundfos product.
	b) No internal communication between the CIU 500 and the Grundfos product.	Check the cable connection between the Grundfos product and CIU 500.
	c) No GRM driver is loaded.	Connect to the GRM server and run the four-step wizard.
4. LED1 is flashing green.	a) 120 seconds startup phase.	
5. LED1 is permanently red.	a) SW1 is in illegal position.	Set the switch to "4".
6. LED2 is permanently red.	a) The module does not support the connected Grundfos product.	Contact the nearest Grundfos company.
7. LED1 is flashing red.	a) The GRM communication is lost or there is no communication to the GRM server.	• Check the Ethernet cable in the router. • Check the router settings. • Check the router related to the SIM card. • Check that the contract with Grundfos is valid and there is access to the GRM data.
8. Cannot access CIU 500 on the GRM interface.	a) No connection to the GRM server.	Check that you use the lowest MAC address (lowest number) of the two ports of CIM 500. See the sticker on the module. Requires a contract with Grundfos and granted access to GRM.

* Requires a contract with Grundfos.

7. Technical data

General	
Application layer	DHCP, HTTP, HTTPS, Ping
Transport layer	TCP, UDP
Internet layer	Internet protocol V4 (IPv4)
Link layer	ARP, Media Access Control, Ethernet
Ethernet cable	Screened, twisted-pair cables, CAT5, CAT5e or CAT6. Auto-crossover detection (auto MDI-X).
Transmission speed	10 Mbit/s, 100 Mbit/s (auto-detected)
Industrial Ethernet protocols	• PROFINET IO • Modbus TCP • BACnet IP • EtherNet/IP • GRM IP*
Supply voltage	5 VDC ± 5 %, $I_{max.}$ 350 mA
Storage temperature	-25 to +70 °C -13 to +158 °F

* Requires a contract with Grundfos.

7.1 PROFINET technical specifications

Functionality PROFINET RT	• PROFINET device according to Conformance Class B • Media Redundancy Protocol (MRP) Client • System redundancy S2 • Multicast provider and subscriber
Minimum cycle time PROFINET RT	250 µs
Number of IO connections per controller	2 for cyclic data 1 for parameter set
Maximum number of IO data	1,024 bytes
GSDML version	V2.34
Dynamic IO configuration	Supported
Diagnostics	Supported
Maximum number of data modules	85
Watchdog	Communication watchdog with fixed 2 seconds time-out. It can be enabled via the control module in the device profile.
Certificate	Conformance July 2019

7.2 Modbus TCP technical specifications

Number of IO socket connections	8
Maximum number of IO data	255 bytes per telegram
Function codes supported	03 Read holding registers 04 Read input registers 06 Write single registers 16 Write multiple registers
Diagnostics	No
DHCP	Supported
Watchdog	Communication watchdog with fixed 5 seconds time-out. It can be enabled via the watchdog register in the device profile.
Certificate	No

To optimise data security when using Modbus TCP via a cellular router, Grundfos strongly recommends that the cellular data connection is based on a private APN with static IP and with no access to public Internet.

7.3 BACnet IP technical specifications

Number of IO socket connections	1
Communication	User Datagram Protocol, UDP
Maximum number of IO data	1500 Bytes
Objects supported	• Analog input • Analog output • Analog value • Binary input • Binary output • Multistate input • Multistate output • Device
DHCP	Supported
Foreign device	Supported
Data sharing services	• ReadProperty • ReadPropertyMultiple • WriteProperty • WritePropertyMultiple • SubscribeCOV • ConfirmedCOVNotification • UnconfirmedCOVNotification
Device management services	• Who-is / I-am • Who-has / I-have • DeviceCommunicationControl
Watchdog	Network watchdog timer. Time-out is configurable via the CIM 500 web page.
Certificate	BTL listing

7.4 EtherNet/IP technical specifications

Minimum requested packet interval	15 ms
I/O data	505 bytes output 509 bytes input Maximum 255 bytes I/O data per assembly.
Number of IO connections	10 Default; configurable depending on available socket resources.
Number of encapsulation sessions	10 Default; configurable depending on available socket resources.
Number of explicit messaging connections	2 explicit messaging connections per encapsulation session 20 explicit messaging connections in total, configurable.
User-specific objects	Object 100. Depending on connected product. • Grundfos pump • Grundfos booster • Grundfos dosing.
Maximum number of connections	2 explicit messaging connections x 10 encapsulation sessions Additional 10 I/O connections Total: 30 connections.
Standard objects	• Identity object (class 0x01) • Message Router object (class 0x02) • Assembly object (class 0x04). Assembly: up to 32 • Connection Manager object (class 0x06) • Device Level Ring (DLR) object (0x47) • Quality of Service (QoS) object (0x48) • TCP/IP Interface object (0xF5) • Ethernet Link object (0xF6)
DHCP	Supported
Functional scope	• Adapter • Support of 2 Ethernet Link objects for implementing ring and daisy chain topologies • Device Level Ring (DLR) protocol (announce-based ring node) • Quality of Service (QoS) • IPv4 Address Conflict Detection (ACD)
Watchdog	Communication watchdog with fixed 5 seconds time-out. It can be enabled via the CIM 500 web page.
Certificate	Plugfest December 2018, Conformance July 2019.

8. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

See also end-of-life information at www.grundfos.com/product-recycling.

Webserver configuration

The built-in webserver offers easy monitoring of the CIM 500 module and makes it possible to configure the selected Industrial Ethernet protocol. Using the webserver, you can also update the firmware of the CIM 500 module and store or restore settings, among other functions.

To connect a PC to CIM 500, proceed as follows:

1. Connect the PC and the module using an ethernet cable.
2. Configure the ethernet port of the PC to the same subnetwork as CIM 500, for example 192.168.1.101. See section *How to configure an IP address on your PC using Windows 7* or *How to configure an IP address on your PC using Windows 10*.
3. Open a standard Internet browser and type 192.168.1.100 in the URL field.

How to configure an IP address on your PC using Windows 7

1. Open "Control Panel".
2. Select "Network and Sharing Center".
3. Click [Change adapter settings].
4. Right-click and select "Properties" for the Ethernet adapter. Typically "Local Area Connection".
5. Select properties for "Internet Protocol Version 4 (TCP/IPv4)".
6. Select the "Alternate Configuration" tab and enter the user-configured IP address and the subnet mask you would like to assign to your PC. See fig. 1.

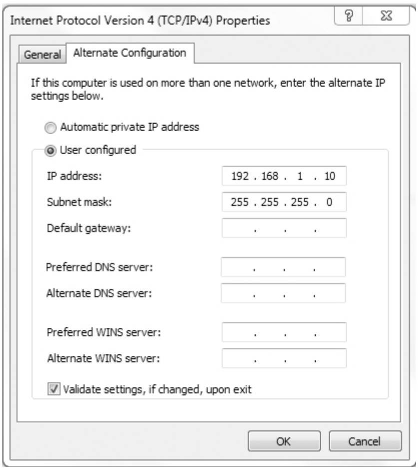

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Fig. 1 Example from Windows 7

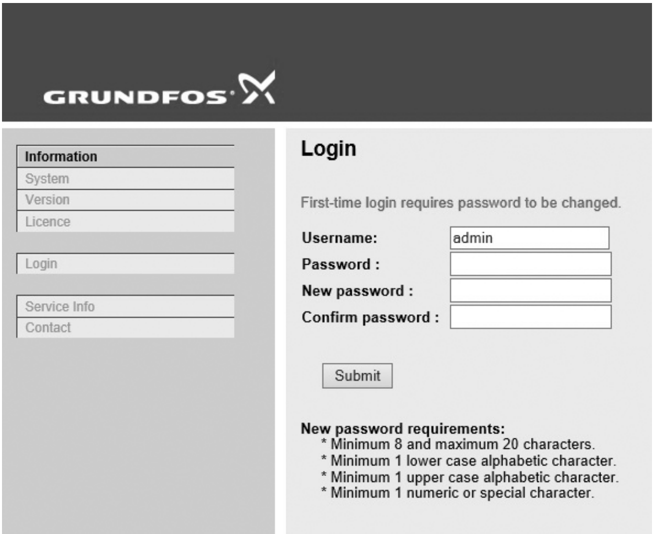
How to configure an IP address on your PC using Windows 10

1. Search for "Ethernet" in Windows.
2. Select "Change Ethernet setting".
3. Select "Change adapter options".
4. Right-click "Ethernet" and select "Properties".
5. Select properties for "Internet Protocol Version 4 (TCP/IPv4)".
6. Select the "Alternate Configuration" tab and enter the user-configured IP address and subnet mask you would like to assign to your PC.

Login

For administration of username and password, see also *User Management*.

If you experience problems with logging in to the webserver after a firmware update, perform a factory reset.



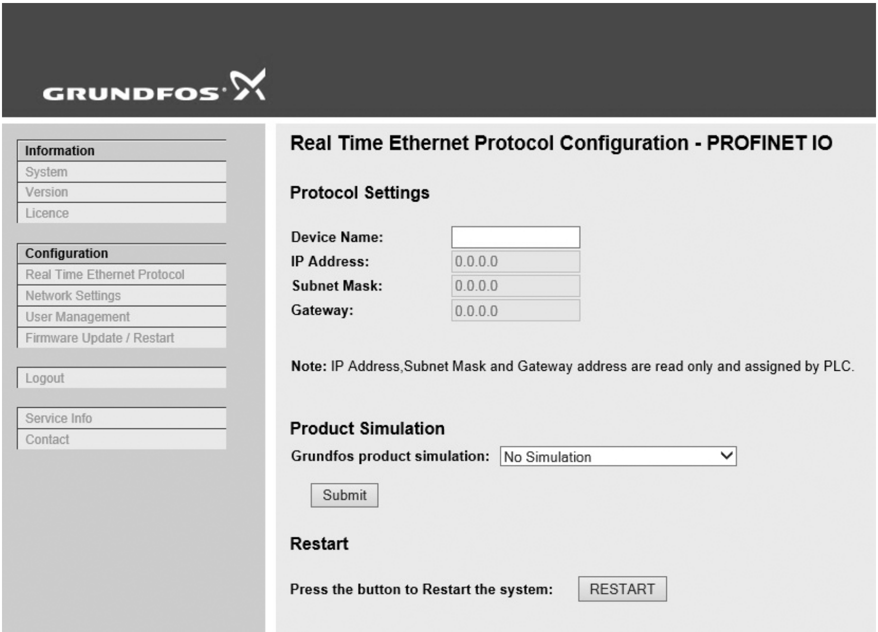
TM07 4522 1919

Fig. 2 Login

Object	Description
Username	Enter username. Default: admin.
Password	Enter password. Default: Grundfos. After the first log in, you are forced to change the password. The password must contain: • at least 8 and maximum 20 characters • at least one lower case letter • at least one upper case letter • at least one numeric or special character. When logging in, you have four attempts before a back-off algorithm starts an exponentially increasing time delay between each attempt. Power cycling CIM 500 resets the back-off algorithm.

PROFINET IO configuration

This web page is used to configure all the parameters relevant to the PROFINET IO protocol standard.



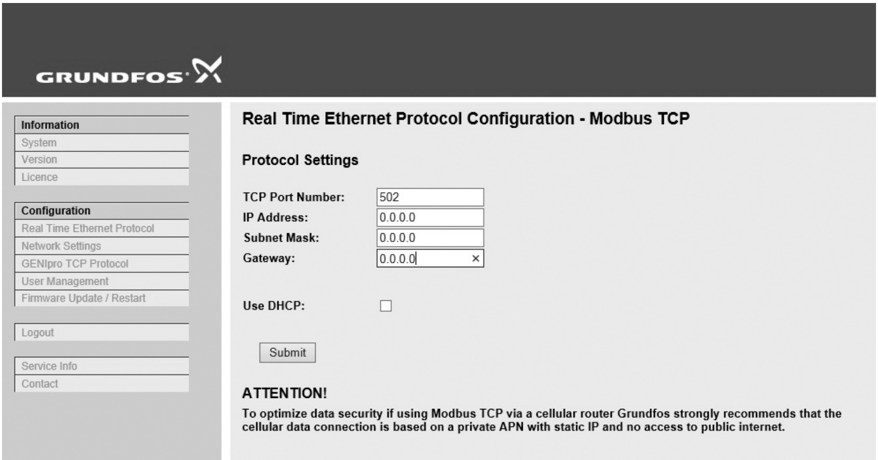
TM07 4525 1919

Fig. 3 RealTime Ethernet Protocol Configuration - PROFINET IO

Object	Description
Device Name	Fill in a device name according to the PROFINET rules: • It consists of one or more labels separated by a dot. • The total length is 1 to 240 characters. • The length of a label is 1 to 63 characters. • Labels only consist of lower case characters, digits and "-". The first and the last character of a label must not be "-".
IP Address	This field is read only. The IP address is assigned from the PLC. PROFINET IO is not allowed to share the IP address with a CIM 500 webserver.
Subnet Mask	This field is read only. The subnet mask is assigned from the PLC.
Gateway	This field is read only. The gateway address is assigned from the PLC.
Grundfos product simulation	The module can be put in product simulation mode to generate realistic simulated values of all the PROFINET IO input data. It will thus be possible to connect a PROFINET IO master to a module fitted in a CIU or an E-box without installing this device in a real industrial process system. In an office environment, it can then be verified that communication works, and data is received and handled correctly by the PROFINET IO master application program (for example PLC program) before installing the device. To enable product simulation, select a product type from the dropdown list. To terminate product simulation, select "No Simulation".
Restart	Press the restart button if the LED1 flashes red, which indicates a wrong or missing PROFINET IO configuration.

Modbus TCP configuration

This web page is used to configure all the parameters relevant to the Modbus TCP protocol standard.



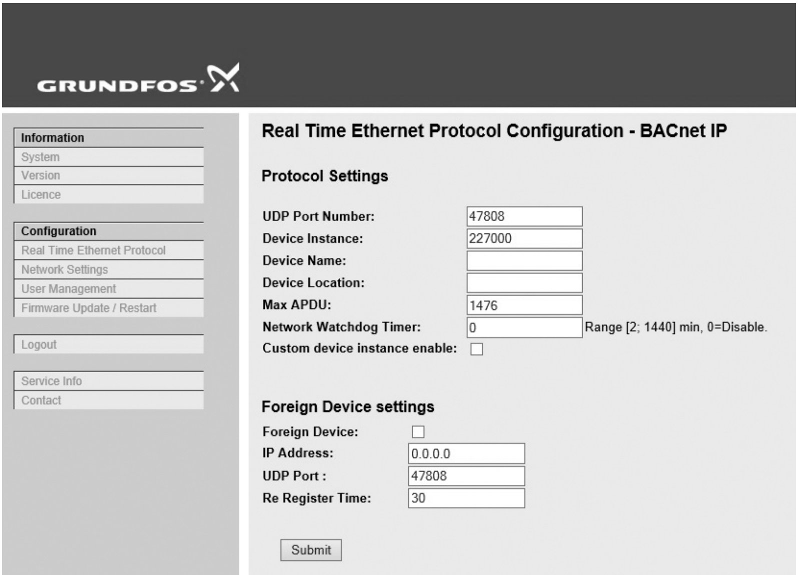
TM07 4523 1919

Fig. 4 Real Time Ethernet Protocol Configuration - Modbus TCP

Object	Description
TCP Port Number	The default value is 502, the official IANA-assigned Modbus TCP port number. The number 502 is always active implicitly. If you select another value in the webserver configuration field, both the new value and value 502 will be active.
IP Address	Configuration of the static IP address if a DHCP server is not used. Modbus TCP is not allowed to share the IP address with a CIM 500 webserver.
Subnet Mask	Configuration of the subnet mask if a DHCP server is not used.
Gateway	Configuration of the gateway address if a DHCP server is not used.
Use DHCP	The module can be configured to automatically obtain its Modbus TCP network settings from a DHCP server if available on the network. Default: DHCP disabled. "Use DHCP" is unchecked.

BACnet IP configuration

This web page is used to configure some of the parameters relevant to the BACnet IP protocol standard. Apart from these settings, BACnet IP will use the IP address, subnet mask, default gateway and DHCP setting configured in the Network Settings menu.



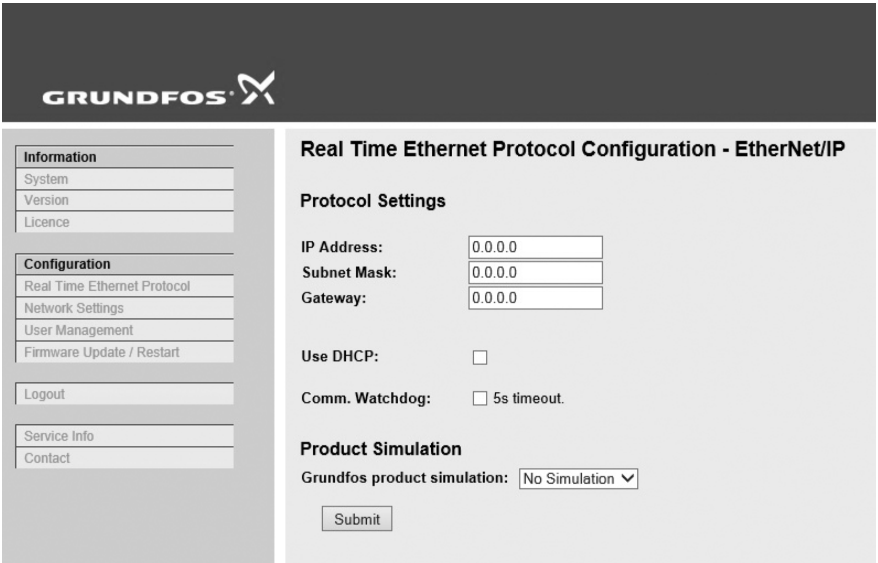
TM07 4518 1919

Fig. 5 Real Time Ethernet Protocol Configuration - BACnet IP

Object	Description
UDP Port Number	Select UDP Port Number. The default number is 47808,the IANA-assigned standard UDP port number for BACnet IP.
Device Instance	Select Device Instance. The default number is 22700. The Device Instance must be unique in the BACnet network. For Grundfos, the number is 227 and it is fixed. The last three digits can be changed.
Device Name	You can name the device. The device name must be unique in the BACnet network. Optional.
Device Location	You can name the device location for local identification. Optional.
Max APDU	Select the maximum Application Protocol Data Unit, between 50 and 1476. The default value is 1476 bytes.
Network Watchdog Timer	If enabled and BACnet communication stops, the watchdog timer restarts the internal network module if it times out. You can configure time-out from 2 minutes to 1440 minutes in steps of 1 minute. A value of 1 is illegal and a value of 0 disables the watchdog.
Custom device instance enable	If ticked, CIM 500 is configured to use a BACnet custom device instance number.
Foreign Device	If ticked, CIM 500 is configured as a foreign device.
IP Address	Enter the foreign IP address.
UDP Port	Select the foreign UDP port number. The default number is 47808.
Re-Register Time	Select the time period in seconds during which the foreign device must re-register on the BACnet network.
Network Watchdog Timer	If enabled and BACnet communication stops, the watchdog timer restarts the internal network module if it times out. You can configure time-out from 2 minutes to 1440 minutes in steps of 1 minute. A value of 1 is illegal and a value of 0 disables the watchdog.

EtherNet/IP configuration

This web page is used to configure all the parameters relevant to the EtherNet/IP protocol standard.



TM07 4519 1919

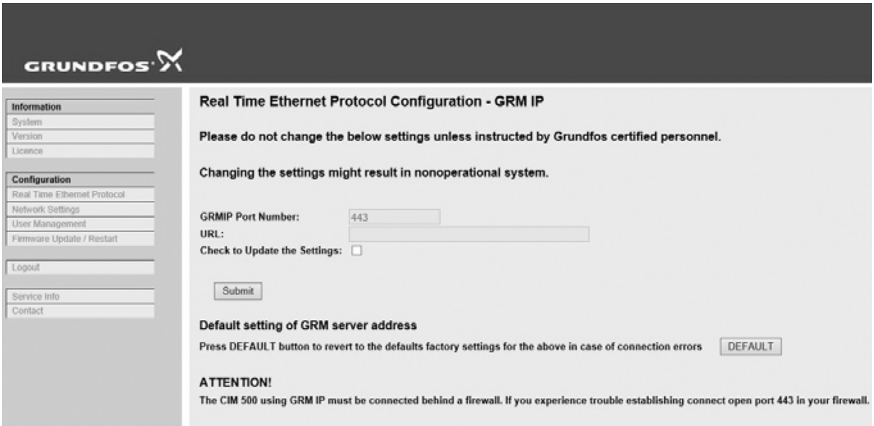
Fig. 6 Real Time Ethernet Protocol Configuration - EtherNet/IP

Object	Description
IP Address	Configuration of the static IP address if a DHCP server is not used. EtherNet/IP is not allowed to share the IP address with a CIM 500 webserver.
Subnet Mask	Configuration of the subnet mask if a DHCP server is not used.
Gateway	Configuration of the gateway address if a DHCP server is not used.
Use DHCP	The CIM 500 module can be configured to automatically obtain its EtherNet/IP network settings from a DHCP server, if available on the network. Default: No use of DHCP.
Communication Watchdog	For enabling of a 5 seconds communication watchdog timer. Only active for pump or booster products. Unchecked: Watchdog is disabled (default) Checked: Watchdog is enabled, time-out 5 seconds. Watchdog action: The pump or the booster is set to local mode.
Grundfos product simulation	The module can be put in product simulation mode to generate realistic simulated values of all the EtherNet/IP input data. It will thus be possible to connect a EtherNet/IP master to a module fitted in a CIU or an E-box without installing this device in a real industrial process system. In an office environment, it can then be verified that communication works, and data is received and handled correctly by the EtherNet/IP master application program (for example PLC program) before installing the device. To enable product simulation, select a product type from the dropdown list. To terminate product simulation, select "No Simulation".

GRM IP configuration

By default, the GRM server URL and DHCP are activated on the GRM webserver.

You need a contract with Grundfos and an external router with Internet connection to gain access to the GRM server.



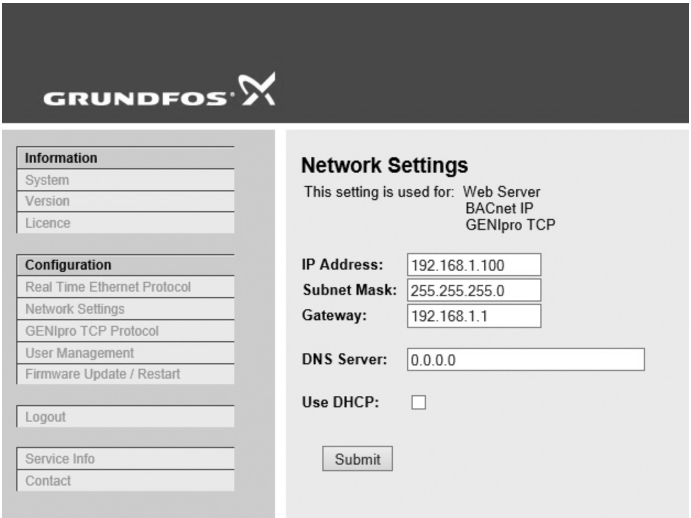
TM07 4521 1919

Fig. 7 Real Time Ethernet Protocol Configuration, GRM IP

Object	Description
GRMIP Port Number	A change is only needed in case Grundfos requires it.
URL	The default URL link to the GRM Server is active but hidden. A change is only needed in case Grundfos requires it.
Check to Update the Settings	This setting enables editing of the URL. Checked: Editing of the URL is possible (not recommended). Unchecked: No editing of the URL is possible (default).
Default setting of GRM server address	Set the URL link to GRM server to default value. Only needed if a manual URL was typed in and should be changed back.

Network Settings

This web page is used to configure the network settings of the webserver and of the GENIpro TCP protocol. The network settings here are also used for BACnet IP. Additional configuration of BACnet IP is done in the Real Time Ethernet Protocol menu. See *EtherNet/IP configuration*.



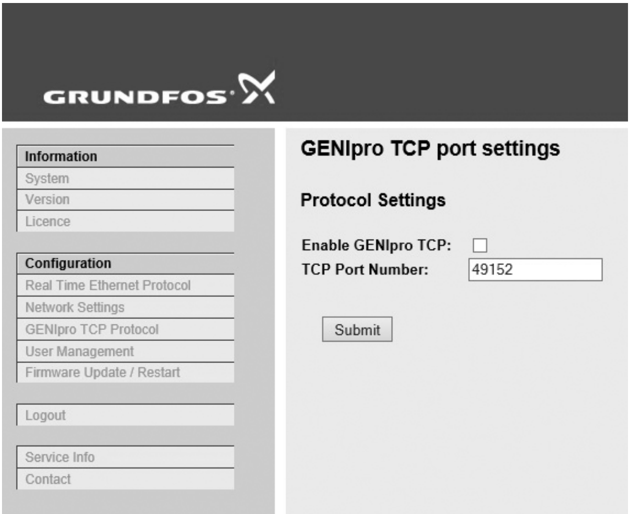
TM07 4524 1919

Fig. 8 Network settings

Object	Description
IP Address	Configuration of the static IP address if a DHCP server is not used. Default: 192.168.1.100.
Subnet Mask	Configuration of the subnet mask if s DHCP server is not used. Default: 255.255.255.0.
Gateway	Configuration of the gateway address if s DHCP server is not used. Default: 192.168.1.1.
DNS Server	The module can be configured to use a specific domain name server, if available on the network. Default: 0.0.0.0.
Use DHCP	The module can be configured to automatically obtain the IP address from a DHCP server, if available on the network. Default: Do not use DHCP.

GENIpro TCP port settings

This web page is only present when the Modbus TCP protocol has been selected, SW1 is in position 1. The module can connect to a Grundfos PC tool and communicate with the product via the GENIpro TCP protocol.



TM07 4520 1919

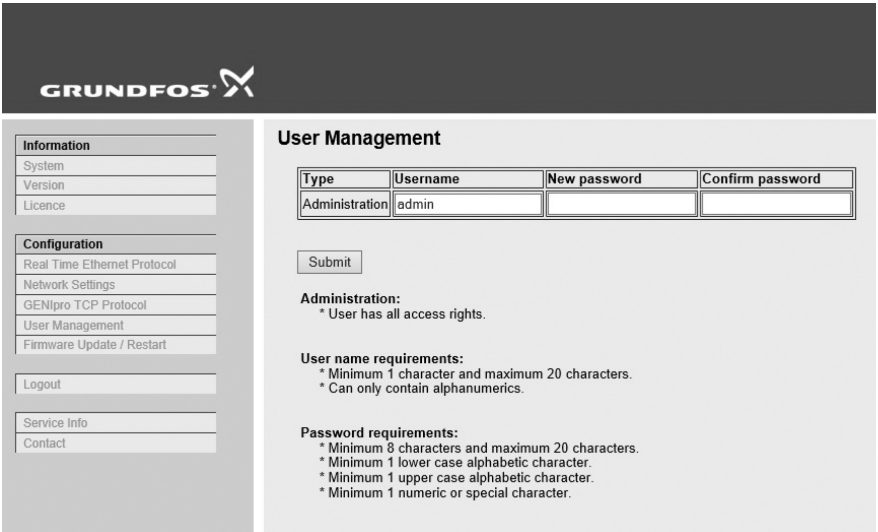
Fig. 9 GENIpro TCP port settings

Object	Description
Enable GENIpro TCP	For enabling or disabling of the GENIpro TCP protocol. Default: Disabled.
TCP Port Number	The TCP port number of the module and that of the PC tool must be identical. Default: 49152, IANA public port.

User Management

A login is required for any change of the CIM 500 settings, and this web page is used to configure the username and password. See *Login*.

It is only possible to configure one user.



TM07 4527 1919

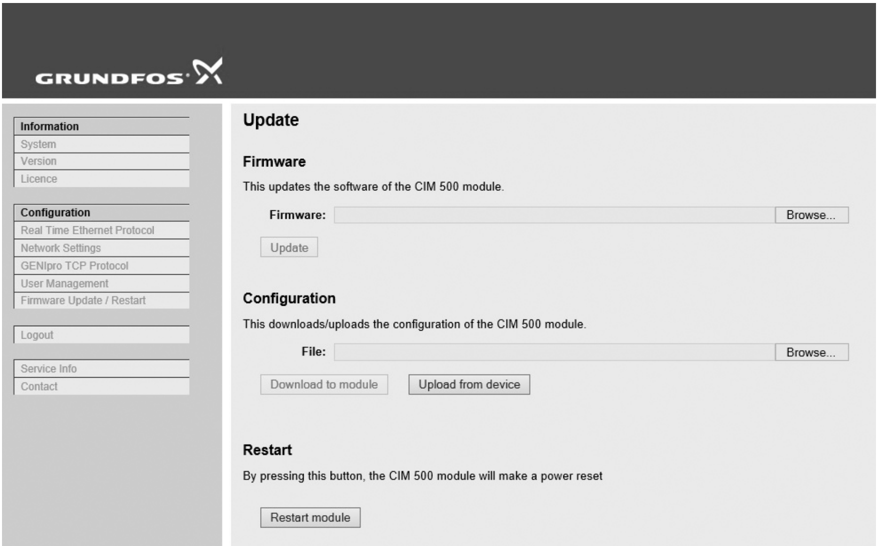
Fig. 10 User management

Update

You can update the firmware by means of the built-in webserver. The binary file is supplied by Grundfos.

To make installation and configuration easier, you can upload the configuration to a PC for backup or distribution to multiple modules.

If you experience problems with logging in to the webserver after a firmware update, perform a factory reset.



TM07 4526 1919

Fig. 11 Update

Object	Description
Firmware	Path to binary firmware image that can be used for updating the module.
Update	Click [Update] to start the update. The procedure takes approximately one minute.
File	Path to the configuration file.
Download to module	Click here to transfer the configuration file to the module.
Upload from device	Click here to upload the configuration of the module to a file on your PC.
Restart module	By pressing this button, the CIM 500 module makes a power-up reset.

中国 RoHS

产品中有害物质的名称及含量

部件名称	有害物质					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr6+)	多溴联苯 (PBB)	多溴联苯醚 (PBDE)
印刷电路板	X	O	O	O	O	O
紧固件	X	O	O	O	O	O

本表格依据 SJ/T 11364 的规定编制

O：表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。
X： 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 该规定的限量要求。

该产品环保使用期限为 10 年，标识如左图所示。
此环保期限只适用于产品在安装与使用说明书中所规定的条件下工作

GB: EC/EU declaration of conformity
We, Grundfos, declare under our sole responsibility that the product CIM 500, to which the declaration below relates, is in conformity with the Council Directives listed below on the approximation of the laws of the EC/EU member states.

DE: EG-/EU-Konformitätserklärung
Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt CIM 500, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-/EU-Mitgliedsstaaten übereinstimmt.

DK: EF-/EU-overensstemmelseserklæring
Vi, Grundfos, erklærer under ansvar at produktet CIM 500 som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EF-/EU-medlemsstaternes lovgivning.

ES: Declaración de conformidad de la CE/UE
Grundfos declara, bajo su exclusiva responsabilidad, que el producto CIM 500 al que hace referencia la siguiente declaración cumple lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la CE/UE.

FI: EY-/EU-vaatimustenmukaisuusvakuutus
Grundfos vakuuttaa omalla vastuullaan, että tuote CIM 500, jota tämä vakuutus koskee, on EY-/EU:n jäsenvaltioiden lainsäädännön lähentämiseen tähtäävien Euroopan neuvoston direktiivien vaatimusten mukainen seuraavasti.

FR: Déclaration de conformité CE/UE
Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit CIM 500, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des États membres CE/UE relatives aux normes énoncées ci-dessous.

IT: Dichiarazione di conformità CE/UE
Grundfos dichiara sotto la sua esclusiva responsabilità che il prodotto CIM 500, al quale si riferisce questa dichiarazione, è conforme alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE/UE.

PT: Declaração de conformidade CE/UE
A Grundfos declara sob sua única responsabilidade que o produto CIM 500, ao qual diz respeito a declaração abaixo, está em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da CE/UE.

RO: Declaraţia de conformitate CE/UE
Noi Grundfos declarăm pe propria răspundere că produsul CIM 500, la care se referă această declaraţie, este în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre CE/UE.

RU: Декларация о соответствии нормам ЕЭС/ЕС
Мы, компания Grundfos, со всей ответственностью заявляем, что изделие CIM 500, к которому относится нижеприведённая декларация, соответствует нижеприведённым Директивам Совета Евросоюза о тождественности законов стран-членов ЕЭС/ЕС.

CN: 欧盟符合性声明
我们，格兰富，在我们的全权责任下声明，产品 CIM 500 系列，其制造和性能完全符合以下所列欧盟委员会指令。

JP: EC/EU 適合宣言
Grundfos は、その責任の下に、CIM 500 製品が EC/EU 加盟諸国の法規に関連する、以下の評議会指令に適合していることを宣言します。

KO: EC/EU 적합성 선언
Grundfos 는 아래의 선언과 관련된 CIM 500 제품이 EU 회원국 법률에 기반하여 아래의 이사회 지침을 준수함을 단독 책임 하에 선언합니다 .

- Low Voltage Directive (2014/35/EU).
  Standard used:
  EN 61010-1:2010
- EMC Directive (2014/30/EU).
  Standards used:
  EN 61000-6-2:2005
  EN 61000-6-4:2007/A1:2011
- RoHS 2 Directive (2011/65/EU and 2015/863/EU)
  Standard used:
  EN 50581:2012

This EC/EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (98407037).

Bjerringbro, 1st September 2019

A. Windfeld

Anne Katrine Windfeld
Senior Manager
Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Manufacturer and person empowered to sign the EC/EU declaration of conformity.

RUS

CIM 500

Руководство по эксплуатации
Руководство по эксплуатации на данное изделие является составным и включает в себя несколько частей:
Часть 1: настоящее «Руководство по эксплуатации».
Часть 2: электронная часть «Паспорт. Руководство по монтажу и эксплуатации» размещенная на сайте компании Грундфос. Перейдите по ссылке, указанной в конце документа.
Часть 3: информация о сроке изготовления, размещенная на фирменной табличке изделия.
Сведения о подтверждении соответствия:
Модули передачи данных CIM 500 прошли процедуру подтверждения на соответствие требованиям Технических регламентов Таможенного союза: ТР ТС 004/2011 «О безопасности низковольтного оборудования»; ТР ТС 020/2011 «Электромагнитная совместимость технических средств».

KAZ

CIM 500

Пайдалану бойынша нұсқаулық
Атаулы өнімге арналған пайдалану бойынша нұсқаулық құрамалы болып келеді және келесі бөлімдерден тұрады:
1 бөлім: атаулы «Пайдалану бойынша нұсқаулық»
2 бөлім: Грундфос компаниясының сайтында орналасқан электронды бөлім «Төлқұжат, Құрастыру және пайдалану бойынша нұсқаулық». Құжат соңында көрсетілген сілтеме арқылы өтіңіз.
3 бөлім: өнімнің фирмалық тақташасында орналасқан шығарылған уақыты жөніндегі мәлімет
Сәйкестік мәлімдемесі туралы ақпарат:
CIM 500 деректерді беру модульдері Кеден одағының техникалық регламенттерінің талаптарына сәйкестігін растау рәсімінен өтті: «Төмен вольтты жабдықтардың қауіпсіздігі туралы» (ТР ТС 004/2011), «Техникалық заттардың электрлі магниттік сәйкестілігі» (ТР ТС 020/2011).

KG

CIM 500

Пайдалануу боюнча колдонмо
Аталган жабдууну пайдалануу боюнча колдонмо курамдык жана өзүнө бир нече бөлүкчөнү камтыйт:
1-Бөлүк: «Пайдалануу боюнча колдонмо»
2-Бөлүк: «Паспорт. Пайдалануу жана монтаж боюнча колдонмо» электрондук бөлүгү Грундфос компаниянын сайтында жайгашкан. Документтин аягында көрсөтүлгөн шилтемеге кайрылыңыз.
3-Бөлүк: жабдуунун фирмалык тактасында жайгашкан даярдоо мөөнөтү тууралуу маалымат.
Шайкештикти баалоо боюнча маалымат алуу үчүн:
Берилмелер модулдар CIM 500 бажы биримдигинин техникалык регламенттерге ылайык текшерилсе: ТР ТБ 004/2011 «Төмөн вольттук жабдуунун коопсуздугу жөнүндө»; ТР ТБ 020/2011 «Техникалык каражаттардын электрмагниттик шайкештиги».

ARM

CIM 500

Շահագործման ձեռնարկ
Տվյալ սարքավորման շահագործման ձեռնարկը բաղկացած է մի քանի մասերից.
Մաս 1. սույն «Շահագործման ձեռնարկ»:
Մաս 2. էլեկտրոնային մաս. այն է` «Անձնագիր: Մոնտաժման և
շահագործման ձեռնարկ» տեղադրված «Գրունդֆոս». Անցեք փաստաթղթի վերջում նշված հղումով.
Մաս 3. տեղեկություն արտադրման ամսաթվի վերաբերյալ` նշված սարքավորման պիտակի վրա:

Համապատասխանության մասին հայտարարության տեղեկություններ՝
CIM 500 տվյալների փոխանցման մոդուլները անցան հաստատման կարգը ՝Մաքսային միության տեխնիկական կանոնակարգի պահանջներին համապատասխան. TR TS 004/2011« voltageածր լարման սարքավորումների անվտանգության մասին »; TR TS 020/2011 «Տեխնիկական միջոցների էլեկտրամագնիսական համատեղելիություն»:

http://net.grundfos.com/qr/i/99800478

10000274311	0120
ECM: 1279071	

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